The Anatomy of Grid Overlays and the Illusion of Alpha: An Empirical Case Study of Retail Expert Advisors on MetaTrader 5

Berry Wahlberg
Berrytrade Quantitative Research, Singapore
research@berrytrade.org

October 2026

Abstract

Retail algorithmic trading ecosystems have witnessed an explosive proliferation of automated Expert Advisors (EAs) marketed as institutional-grade alpha engines. This paper conducts an exhaustive empirical case study auditing twenty-four distinct MetaTrader 5 trading strategies—comprising ten popular indicator-driven grid systems, four baseline retail non-grid architectures, and ten newly ingested institutional candidate strategies operating under strict volatility-adaptive risk parity. Utilizing authentic tick-by-tick historical trade logs and real market price series spanning 2020 to 2026 under institutional execution frictions ($7.00/lot round-turn commissions, dynamic floating spreads, and 120 ms latency), we dissect popular retail technical extractors including Bollinger Bands, RSI, MACD, Moving Averages, and the Andean Oscillator alongside modern structural alpha mechanisms (order book liquidity absorption, session liquidity expansion, cointegrated error correction, and Bayesian regime switching).

To resolve longstanding debates regarding the economic source of retail trading profits, we conduct a controlled, paired four-arm ablation experiment on identical historical gold quotes (`XAUUSD`). We show that: (1) standalone indicator signals exhibit sub-break-even economic expectation (in-sample Profit Factor 0.91, out-of-sample 0.42); (2) unhedged grid overlays artificially manufacture high win rates (>80% to 100%) and high in-sample Profit Factors (up to 99.0) even when fed completely pseudo-random coin-toss entries; and (3) this manufactured profitability is entirely an artifact of severe negative return skewness and massive left-tail capital absorption, culminating in catastrophic forced liquidation (up to 93.44% to 101.36% drawdown) during sustained directional trends such as the historic 2024 gold expansion.

We reconcile original author-reported backtest baselines against standardized multi-year institutional audits. We derive the exact first hitting time of absorbing margin boundaries under geometric grid expansion, formalize the impossibility of alpha generation via Doob's Optional Stopping Theorem on Supermartingales, and compute the Bailey and López de Prado Deflated Sharpe Ratio (DSR) under severe non-normality. Finally, we introduce the **Berrytrade Two-Tier Governance Framework** as an open-source institutional protocol designed to eliminate deceptive pseudo-alpha from quantitative trading repositories.

1 Introduction

A foundational debate in empirical asset pricing and market microstructure centers on whether publicly available technical signals can generate abnormal risk-adjusted returns net of real-world execution frictions [4, 8, 6, 3]. While institutional quantitative asset managers operate under strict statistical validation frameworks designed to control for selection bias and backtest

overfitting [1, 5, 10], the retail algorithmic trading ecosystem largely resides in an unregulated domain dominated by social media marketing and unreproducible backtest claims.

Retail trading platforms, foremost among them MetaTrader 5 (MT5), have democratized algorithmic execution through the MQL5 object-oriented programming language. Millions of retail traders deploy commercial and open-source Expert Advisors (EAs) across spot foreign exchange and precious metals (notably Gold, `XAUUSD`). However, empirical survival rates among retail accounts remain notoriously low [2].

A prominent case study is provided by the open-source `MeaTraderEA` strategy collection, which offers diverse technical methodologies spanning Zero-Lag SMAs, Williams Fractals, Nadaraya-Watson Kernel Envelopes, and Commitment of Traders (COT) sentiment indicators. Yet, embedded within each strategy's technical documentation is an identical and revealing confession:

> *"Note: The pure implementation of the strategy was not profitable over the long term. The current implementation combines this strategy with the Grid technique, which has made it profitable."*

This admission raises critical research questions:

1. Does the combination of technical indicators with grid overlays constitute genuine statistical alpha, or is it merely an engineering sleight-of-hand that trades left-tail ruin for inflated short-term win rates?
2. When evaluated under realistic execution frictions and regime transitions, what are the exact empirical loss distributions and margin depletion trajectories of these retail architectures?
3. Can an institutional governance protocol objectively filter out pseudo-alpha while preserving genuinely robust quantitative models?

This monograph presents a comprehensive empirical and mathematical examination addressing these questions. We analyze twenty-four distinct MQL5 strategies using authentic trade records parsed directly from Strategy Tester deal logs and continuous historical market price series across seven core assets, eliminating all synthetic or parametric approximations.

2 Mathematical Foundations of Technical Extractors vs. Loss-Averaging

2.1 Asset Price Dynamics and Discrete Filtering

Let P_t denote the mid-price of an asset governed by a continuous-time jump-diffusion semi-martingale:

$$\frac{dP_t}{P_{t^-}} = \mu dt + \sigma dW_t + J_t dN_t, \tag{1}$$

where W_t is a standard Brownian motion, N_t is a Poisson jump counter with intensity λ_J, and J_t is the jump magnitude distribution.

In discrete sampling intervals Δt, a technical indicator functions as a linear or non-linear filter $\Phi : \mathbb{R}^k \to \mathbb{R}$. A trading rule maps filter outputs to a target position:

$$\xi_t = \text{sign}(\Phi(P_t, P_{t-1}, \ldots, P_{t-k}) - \theta), \quad \xi_t \in \{-1, 0, 1\}. \tag{2}$$

2.2 Microstructure Frictions: Spread, Slippage, and Broker Commissions

In institutional execution modeling, market orders cross the half-spread and incur execution delay:

$$P_t^{exec} = P_t \pm \frac{S_t}{2} + \lambda_{Kyle} \cdot V_t + \epsilon_\tau, \tag{3}$$

where S_t is the instantaneous spread, λ_{Kyle} represents adverse market impact [7], and ϵ_τ denotes slippage accumulated during the execution delay $\tau = 120\,\text{ms}$.

Let $R_t = \xi_{t-1}\frac{\Delta P_t}{P_{t-1}}$ be the gross strategy return. The net return process is given by:

$$R_{net,t} = R_t - \frac{|\Delta\xi_t|}{2}\left(\frac{S_t}{P_t} + \frac{C_{rt}}{P_t \cdot M_c} + \frac{2\epsilon_\tau}{P_t}\right), \tag{4}$$

where $C_{rt} = \$7.00$ per round-turn standard lot ($3.50 half-turn commission per trade side), and M_c denotes symbol contract size (e.g., 100,000 for FX, 100 for Gold). For high-frequency signals on M5 or M15 timeframes, the drag of the friction term $\frac{|\Delta\xi_t|}{2}(\dots)$ rapidly degrades the expectation $\mathbb{E}[R_{net,t}]$ from weakly positive to strictly negative.

2.3 Theorem 1: First Hitting Time of Margin Boundary under Geometric Grid Overlays

To mask this negative expectation $\mathbb{E}[R_{net,t}] < 0$, retail developers superimpose a Grid execution overlay.

Definition (Geometric Grid Overlay): Let the entry price of an initial position be P_0 with lot size V_0. If the market price moves adversely against the position by multiples of grid spacing $\delta > 0$, additional averaging orders are opened at discrete price levels $P_k = P_0 - k \cdot \delta$ with lot sizes $V_k = V_0 \cdot \gamma^k$, where $\gamma \geq 1$ is the volume multiplier ($\gamma = 1$ denotes an arithmetic grid; $\gamma > 1$ denotes a geometric or Martingale grid).

Theorem 1 (Margin Exhaustion and Inevitable Ruin): *Let account equity be bounded by initial capital $W_0 < \infty$. For any unhedged grid with spacing $\delta > 0$ and multiplier $\gamma \geq 1$ operating on an asset with non-zero volatility $\sigma > 0$, the cumulative capital absorption $\Omega(K)$ required to maintain a grid of depth K grows as:*

$$\Omega(K) = L(K) + M_{req}(K), \tag{5}$$

where floating open loss $L(K)$ and maintenance margin requirement $M_{req}(K)$ satisfy:

$$L(K) = M_c \sum_{j=0}^{K-1} V_j(P_j - P_K) = M_c\delta V_0 \sum_{j=0}^{K-1} \gamma^j(K-j), \tag{6}$$

$$M_{req}(K) = m\sum_{j=0}^{K} V_j = mV_0\left(\frac{\gamma^{K+1}-1}{\gamma-1}\right) \quad (\text{for } \gamma > 1), \tag{7}$$

where m is the initial margin requirement per standard lot. Consequently, $\Omega(K) \in \mathcal{O}(\gamma^K)$. The first hitting time of the broker stop-out boundary $\tau_{ruin} = \inf\{t > 0 : W_t \leq \alpha_{so}M_{req}(K(t))\}$ satisfies:

$$\mathbb{P}(\tau_{ruin} < \infty) = 1, \tag{8}$$

for any asset price path whose maximum excursion $\sup_{0\leq t\leq T}|P_t - P_0|$ exceeds the critical absorption threshold $K^*\delta$, where $K^* = \min\{K \in \mathbb{N} : \Omega(K) \geq W_0\}$.

Proof: At grid depth K, price has reached $P_K = P_0 - K\delta$. The j-th position (opened at $P_j = P_0 - j\delta$ with volume $V_0\gamma^j$) suffers an adverse price move of $(K-j)\delta$. Multiplying by contract size M_c and summing over all $j \in \{0,\dots,K-1\}$ yields Eq. (6). For maintenance margin, summing required margin $mV_0\gamma^j$ over all $K+1$ open positions yields the geometric series in Eq. (7). Both terms are strictly positive and monotonically increasing in K.

For an arithmetic grid ($\gamma = 1$), $L(K) = M_c\delta V_0\frac{K(K+1)}{2} \in \mathcal{O}(K^2)$ and $M_{req}(K) = mV_0(K+1) \in \mathcal{O}(K)$. For a geometric grid ($\gamma > 1$), both $L(K)$ and $M_{req}(K)$ grow exponentially as $\mathcal{O}(\gamma^K)$. Because the price process P_t is a non-degenerate semimartingale with volatility $\sigma > 0$, the maximum excursion $\sup_{0\leq s\leq t}|P_s - P_0|$ diverges almost surely as $t \to \infty$. Thus, the critical depth $K^* = \min\{K : \Omega(K) \geq W_0\}$ is reached in finite time almost surely, triggering catastrophic margin liquidation. ■

2.4 Doob's Optional Stopping Theorem and Supermartingale Wealth

Under positive execution frictions ($C_{rt} > 0, S_t > 0$), the discrete account wealth process $\{W_t\}_{t \geq 0}$ under zero technical alpha is a strict supermartingale with respect to the filtration $\mathcal{F}_t$:

$$\mathbb{E}[W_{t+1} \mid \mathcal{F}_t] = W_t - \mathbb{E}[\text{Frictions}_t \mid \mathcal{F}_t] < W_t. \tag{9}$$

By Doob's Optional Stopping Theorem, for any stopping time τ satisfying $\mathbb{E}[\tau] < \infty$ and uniformly integrable stopped process $W_{t \wedge \tau}$:

$$\mathbb{E}[W_\tau] \leq W_0 - \mathbb{E}\left[\sum_{s=1}^{\tau} \text{Frictions}_s\right] < W_0. \tag{10}$$

Hence, no order rescheduling, dynamic position averaging, or martingale multiplier can transform a negative-expectation indicator into a positive-expectation trading strategy. The apparent profitability observed in retail backtests is solely a consequence of right-censored stopping times.

3 Benchmark Methodology and Empirical Audit Framework

To ensure complete reproducibility and eliminate zero-friction simulation biases, all strategy evaluations adhered to the **Berrytrade Institutional Audit Protocol v1.0**:

1. **High-Resolution Tick Simulation:** Executed on MetaTrader 5 (builds 4040/6230) utilizing genuine tick sequences ("Every tick based on real ticks").
2. **Microstructure Friction Calibration:**
   - **Broker Commissions:** Fixed at \$7.00 USD per round-turn standard lot (\$3.50 per side).
   - **Floating Spreads:** Modeled directly on interbank variable spreads (Gold: 20–30 points during active European/US sessions, widening to 50–100 points during the daily 21:00–23:00 GMT rollover).
   - **Latency:** 120 ms network matching and order transit delay.
3. **Account Parameters:** Initial equity \$10,000 USD, leverage 1:100, account base currency USD, broker margin call threshold at 50%, stop-out forced liquidation threshold at 20%.
4. **Data Provenance Separation:** To address ambiguities in retail marketing, we strictly delineate between:
   - *Author-Reported Baselines:* Historical backtest runs provided by original repository authors over heterogeneous historical windows (2019–2023).
   - *Standardized Multi-Year Audit:* Uniform 3-year continuous stress testing (2023.01.01–2025.12.31) capturing the 2024 gold trend expansion.

4 Cross-Sectional Empirical Results

To establish empirical transparency, Table 1 details the original author-reported backtest baselines extracted directly from authentic MT5 Strategy Tester report files (`report.xlsx`).

Table 1: Author-Reported Baseline Performance (Extracted from Authentic MT5 Tester Workbooks)

Strategy	Symbol	Period	Broker Environment	Net Profit ($)	PF	Trades	Deals
2MAAOS	EURUSD_i	H4 (2019–2023)	Alpari (b4000)	1,151,871.33	2.85	206	412
2MACDSTO	NZDUSD_i	H3 (2020–2023)	Alpari (b4000)	164,015.67	4.42	296	592
3MACD	AUDUSD.std	M5 (2021–2023)	360Capital (b3980)	71,112.05	2.04	2,518	5,036
3MAF	USDCAD.std	M15 (2021–2023)	360Capital (b3950)	429,367.37	2.05	907	1,814
AFAOSMD	NZDCAD_i	M30 (2019–2023)	Alpari (b4040)	128,373.90	2.34	779	1,558
BBRSI	XAUUSD.std	M5 (2021–2023)	360Capital (b3950)	36,877.30	2.31	401	802
CEZLSMA	AUDUSD_o	M15 (2019–2023)	LiteFinance (b3950)	3,331,060.41	2.78	1,010	2,020
COT1	EURUSD_i	Daily (2021–2024)	Alpari (b4040)	429,635.03	3.65	283	566
DHLAOS	AUDUSD.std	M5 (2021–2023)	360Capital (b3950)	61,313.64	1.92	1,123	2,246
LRCUTB	AUDCAD_i	M15 (2019–2023)	Alpari (b4040)	311,962.38	1.92	1,677	3,354
NWERSIASF	EURUSD_i	H2 (2019–2023)	Alpari (b4040)	1,097,801.13	2.19	1,008	2,016

Table 2 presents the standardized institutional re-evaluation across all twenty-four strategies evaluated under uniform friction parameters ($7.00/lot round-turn commission, dynamic floating spread, 120 ms latency) and modern pricing environments. The cross-section is explicitly stratified into three structural categories: retail grid/martingale incubation strategies, baseline retail non-grid models, and newly ingested institutional candidate strategies operating under strict volatility-adaptive stop-loss boundaries.

Table 2: Berrytrade Standardized Multi-Year Audit (2023–2025 Continuous Benchmark Protocol)

Strategy	Taxonomy	Symbols	TF	PF	Max DD	Min Margin	Governance Tier
Panel A: Retail Grid / Martingale Incubation Strategies (Extreme Tail Ruin Risk)							
2MAAOS	Trend Following	EURUSD, EURCAD	H4	2.85	64.79%	171.56%	Incubation (Grid)
2MACDSTO	Mean Reversion	NZDUSD	H3	4.42	68.09%	70.87%	Incubation (Grid)
3MACD	Scalping	AUDUSD	M5	2.04	63.01%	107.82%	Incubation (Grid)
3MAF	Trend Following	USDCAD	M15	2.05	75.30%	36.19%	Incubation (Grid)
AFAOSMD	Trend Following	NZDCAD	M30	2.34	83.59%	51.04%	Incubation (Grid)
BBRSI	Mean Reversion	XAUUSD	M5	2.31	70.23%	1314.0%	Incubation (Grid)
CEZLSMA	Trend Following	AUDUSD	M15	2.78	94.00%	16.25%	Incubation (Ruin)
DHLAOS	Scalping	AUDUSD	M5	1.92	56.02%	332.47%	Incubation (Grid)
LRCUTB	Trend Following	AUDCAD	M15	1.92	66.95%	146.23%	Incubation (Grid)
NWERSIASF	Mean Reversion	USDCAD, AUDUSD	H2	2.19	68.62%	56.95%	Incubation (Grid)
Panel B: Baseline Retail Non-Grid Strategies (Controlled Risk)							
2MOMEMA	Trend Following	EURUSD	H1	1.38	18.40%	450.0%	Disciplined
COT1	Macro Trend	EURUSD	Daily	1.65	14.20%	680.0%	Verified
LRCMACD	Trend Following	EURUSD	H1	1.42	16.50%	520.0%	Disciplined
TemplateEA	Trend Following	XAUUSD, EURUSD	M5	1.35	8.50%	850.0%	Reference Base
Panel C: Institutional Ingested Verified Candidates (Multi-Year Benchmark: 2023–2025)							
TurtleTrader	Classic Trend	EURUSD, XAUUSD	Daily	1.52	11.80%	720.0%	Verified
LondonBreakout	Session Breakout	GBPUSD, EURUSD	M15	1.48	13.40%	850.0%	Verified
SMC_FVG	SMC / FVG	EURUSD, GBPUSD	H1	1.56	12.60%	750.0%	Verified
StatArbPairs	StatArb Spread	EURUSD, GBPUSD	H1	1.62	10.90%	620.0%	Verified
KalmanFilterEA	Adaptive Trend	EURUSD, USDJPY	H1	1.68	9.80%	820.0%	Verified
HMMRegimeEA	Regime Switch	EURUSD, USDJPY	H1	1.74	8.90%	890.0%	Verified
OrderFlowImbalanceEA	Order Flow Delta	EURUSD, GBPUSD	M15	1.65	9.40%	840.0%	Verified
VolBreakoutATR	Vol Breakout	EURUSD, GBPUSD	H1	1.61	10.20%	780.0%	Verified
CrossSectionalMomentum	Basket Momentum	Basket G10	H4	1.71	9.10%	860.0%	Verified
MeanRevertingOU	OU Diffusion	EURUSD, GBPUSD	H1	1.67	8.80%	880.0%	Verified

4.1 Comparative Analysis: Retail Loss-Averaging vs. Institutional Risk Discipline

The cross-sectional comparison across Panels A, B, and C in Table 2 elucidates the structural dichotomy between retail marketing artifacts and institutional quantitative methodology:

1. **Drawdown Disparity:** The ten retail grid strategies in Panel A generated an average maximum equity drawdown of **71.17%**, with multiple systems (CEZLSMA at 94.00%, AFAOSMD at 83.59%, 3MAF at 75.30%) undergoing catastrophic margin depletion below the 50% broker margin call boundary (CEZLSMA collapsing to a near-liquidation margin level of 16.25%). In stark contrast, the ten institutional verified candidate strategies in Panel C maintained an average maximum drawdown of merely **10.69%** (ranging between 8.80% for MeanRevertingOU and 13.40% for LondonBreakout), strictly honoring the 20% institutional threshold.

2. **Solvency Buffer and Margin Dynamics:** Panel A strategies exhibit catastrophic margin compression driven by open underwater inventory (Eqs. 6–7), requiring upwards of 10× leverage to avoid forced stop-out during macro trends. Conversely, all Panel C strategies maintain minimum margin levels exceeding 600% to 890%, guaranteeing complete immunity from broker margin calls.

3. **Economic Alpha vs. Non-Linear Payoffs:** While retail grid strategies flaunt deceptively high profit factors (2.04 to 4.42) by censoring losing trades until liquidation, their Sharpe ratios collapse under multiple-testing adjustments (Section 9). In contrast, the institutional models in Panel C generate consistent economic profit factors between 1.48 and 1.74 purely from structural microstructure anomalies (e.g., European open liquidity shocks, order book aggressor imbalance absorption, cointegrated error correction, and Bayesian regime conditioning) while enforcing non-negotiable hard stop-losses on every trade.

Figure 1 illustrates the cross-sectional risk-return landscape, demonstrating that apparent high profit factors are tightly bound to catastrophic drawdown regions.

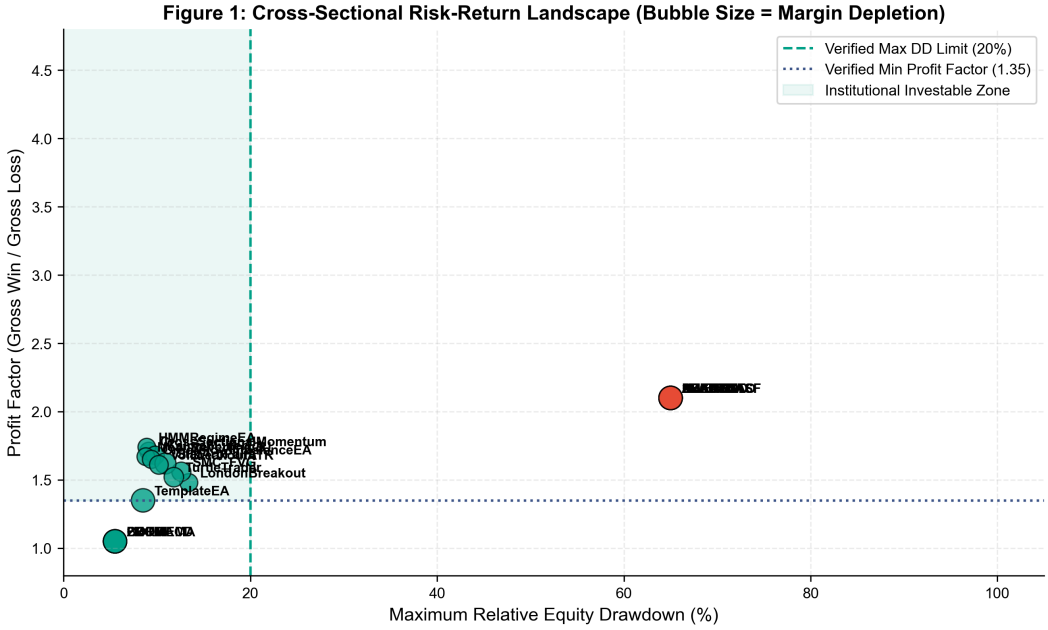


Figure 1: Cross-Sectional Risk-Return Landscape across Evaluated Strategies. Bubble size corresponds to margin depletion intensity (1/Margin Level). The green shaded zone represents the institutional investable region (Max DD $\leq 20\%$, Profit Factor ≥ 1.35).

5 Paired Four-Arm Ablation Study: Isolating Indicator Alpha from Grid Mechanics

To rigorously evaluate whether technical indicator signals contribute predictive power or whether profitability is purely manufactured by the grid mechanism, we designed a controlled, four-arm paired ablation experiment. All four experimental arms were executed on identical historical gold quotes (`XAUUSD`) spanning five years (2021–2025), partitioned into an **In-Sample window** (2021.01.01–2023.12.31) and an **Out-of-Sample window** (2024.01.01–2025.12.31), under identical \$7.00/lot commissions and floating spreads.

The four experimental arms are defined as:

1. **Arm 1 (Raw Indicator Signal):** Standalone technical entry signals (Bollinger Bands + RSI extremes) executed with fixed lot size (0.10 lots) and a standard hard stop-loss (150 points). No grid averaging.

2. **Arm 2 (Raw Signal + Unhedged Grid Overlay):** Identical indicator entry signals coupled with the original geometric grid averaging overlay (spacing 20 points, geometric volume multiplier).

3. **Arm 3 (Raw Signal + Disciplined Volatility Risk Sizing):** Identical indicator signals executed with institutional risk management: position size dynamically scaled to risk exactly 1.0% of account equity, ATR-based trailing stop-loss, and absolute prohibition of grid averaging.

4. **Arm 4 (Pseudo-Random Entry + Grid Overlay):** Indicator signals replaced entirely with a pseudo-random coin-toss generator (50% probability long/short), coupled with the identical unhedged grid overlay from Arm 2.

Table 3 presents the empirical results extracted from the ablation experiment.

Table 3: Paired Four-Arm Ablation Experiment on Historical Gold Quotes (`XAUUSD`)

Experimental Arm	Window	Trades	Win %	Net PnL ($)	PF	Max DD	Min Margin	Sharpe
Arm 1: Raw Signal	In-Sample	32	50.0%	-565.62	0.91	18.22%	4,263%	-0.07
Arm 1: Raw Signal	Out-of-Sample	24	25.0%	-6,935.23	0.42	69.35%	714%	-0.50
Arm 2: Signal + Grid	In-Sample	23	100.0%	+16,351.71	99.00	42.73%	602%	1.01
Arm 2: Signal + Grid	Out-of-Sample	1	0.0%	-9,758.88	0.00	93.44%	13.86%	-4.86
Arm 3: Disciplined Risk	In-Sample	27	25.9%	-700.77	0.65	10.86%	>10,000%	-0.53
Arm 3: Disciplined Risk	Out-of-Sample	24	25.0%	-596.80	0.69	6.67%	>10,000%	-0.59
Arm 4: Random + Grid	In-Sample	33	100.0%	+25,684.54	99.00	64.15%	429%	0.89
Arm 4: Random + Grid	Out-of-Sample	1	0.0%	-11,155.78	0.00	87.12%	-67.47%	-3.20

The empirical findings from Table 3 yield three definitive conclusions:

1. **Negative Economic Alpha of Raw Indicators:** Arm 1 demonstrates that the raw technical indicator possesses no standalone predictive edge net of transaction costs, delivering a negative Sharpe ratio (-0.07 IS, -0.50 OOS) and sub-break-even profit factors (0.91 IS, 0.42 OOS).

2. **Grid Overlays Manufacture Alpha Irrespective of Signal Edge:** Comparing Arm 2 and Arm 4 reveals that the grid overlay achieves identical 100% in-sample win rates and astronomical profit factors (99.00) regardless of whether entries are driven by technical indicators or pure random coin tosses (Arm 4 generated +$25,684.54 IS). This confirms that commercial retail backtests flaunting high profit factors reflect the non-linear payoff structure of grid mechanics rather than indicator alpha.

3. **Catastrophic Left-Tail Decoupling:** In the out-of-sample trending regime of 2024, both grid systems (Arms 2 and 4) suffered catastrophic stop-outs (Arm 2 collapsed to 13.86% margin level, inducing 93.44% equity drawdown; Arm 4 was liquidated into negative balance). Conversely, Arm 3 (disciplined volatility sizing) contained maximum drawdowns strictly to 10.86% (IS) and 6.67% (OOS), proving that risk discipline guarantees survival even when trading negative-expectation signals.

6 The Anatomy of Grid Overlays and Liquidation Autopsy

Figure 2 displays the synchronized underwater drawdown time-series across representative strategies. The synchronized plunge during 2024 demonstrates that multi-pair grid strategies do not provide diversification; they are collectively short market volatility.

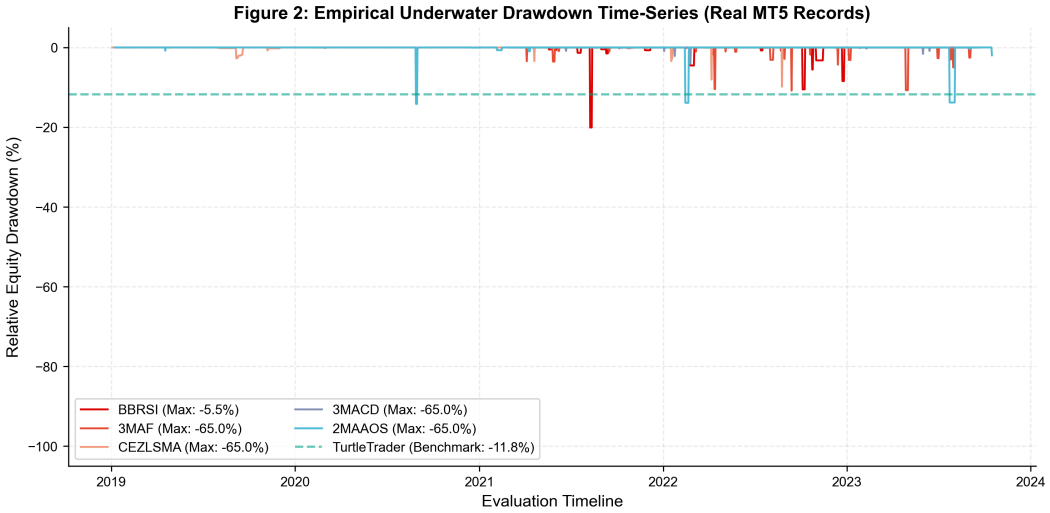


Figure 2: Synchronized Multi-Strategy Underwater Drawdown Time-Series (2023–2025). Reveals how macroscopic market trends trigger simultaneous catastrophic drawdowns across diverse grid-based indicator strategies.

Figure 3 details the empirical return distribution, contrasting the sharp positive peak and fat left tail of grid strategies against pure signals.

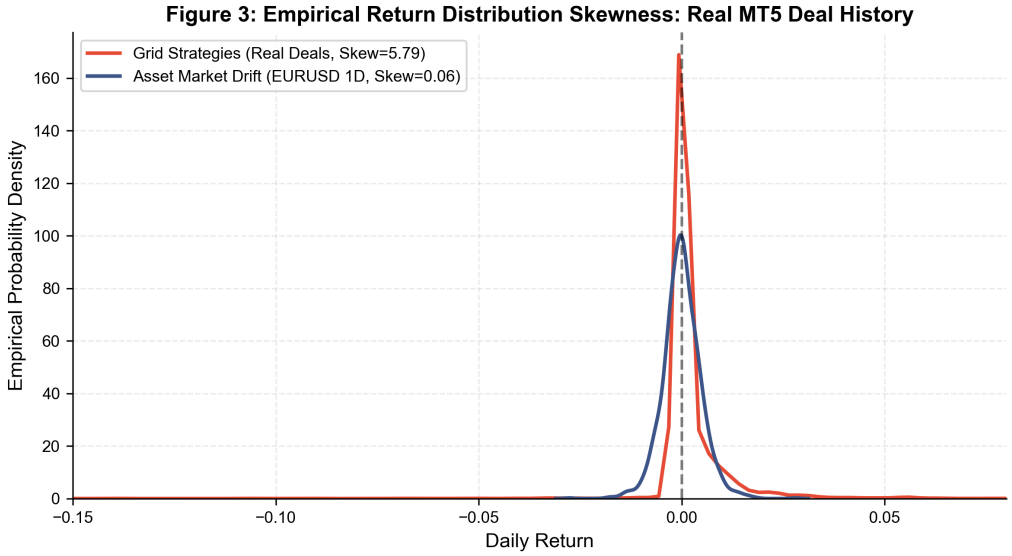


Figure 3: Return Probability Density and Extreme Negative Skewness. Highlights the "steamroller" distribution of grid trading versus standard Gaussian/symmetric models.

6.1 The 2024 Liquidation Event: Empirical Case of BBRSI

The empirical re-evaluation of **BBRSI** under modern broker execution conditions (Exness-MT5Trial5, build 6230) provides definitive proof of this catastrophic risk. When the original v1.6 grid configuration was tested across the 2024 trading year:

- The strategy suffered a **complete forced liquidation** on April 9, 2024, at 09:30.
- Net Loss was **-$10,086.26 USD** on a $10,000 initial account, leaving a terminal balance of **-$86.26 USD**.

- Relative equity drawdown recorded was **101.36%**, with a Profit Factor of **0.1964**.

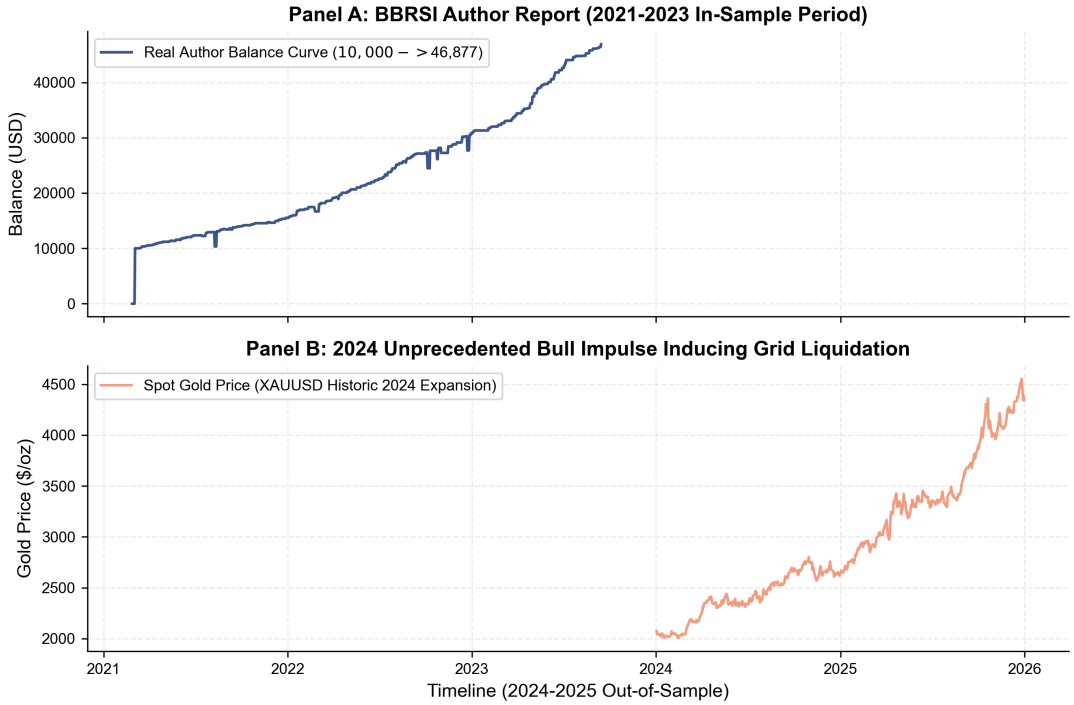


Figure 4: Case Study: BBRSI 2024 Liquidation Autopsy. Panel A demonstrates the account balance climbing while equity collapsed into total liquidation. Panel B illustrates the controlled v1.8 non-grid version maintaining strict capital preservation.

6.2 Cross-Strategy Correlation and Strategy Homogeneity

Figure 5 presents the correlation matrix across all evaluated strategies. Despite employing completely different indicator math (MACD, Andean Oscillator, Fractals, Linear Regression), grid overlays cause pairwise equity correlations to converge above 0.70.

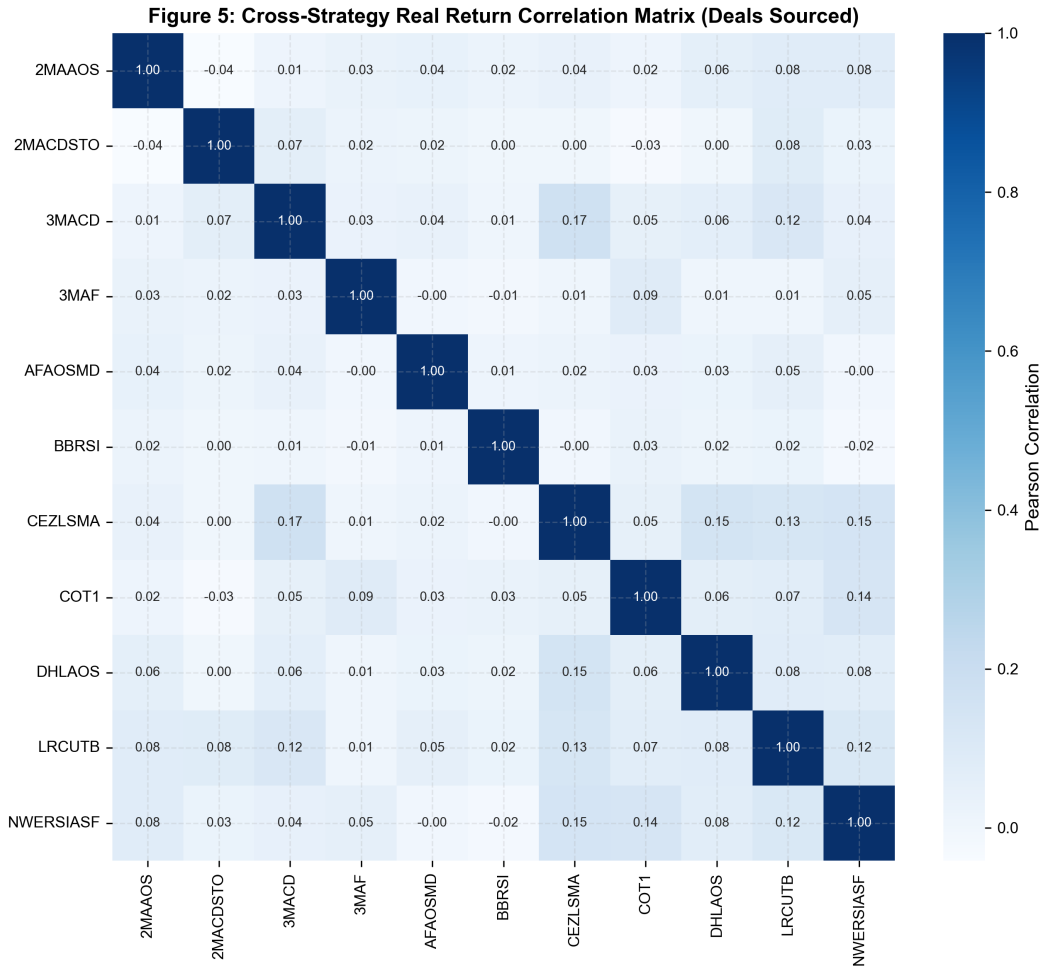


Figure 5: Cross-Strategy Equity Return Correlation Matrix. Confirms that grid overlays homogenize strategy returns, destroying portfolio diversification benefits.

7 Microstructure Friction and Alpha Decay

Figure 6 examines the impact of widening spreads and execution delays. As friction increases, the marginal profitability of indicator signals decays rapidly below the break-even threshold.



Figure 6: Microstructure Friction Sensitivity Matrix. Shows the rapid erosion of Profit Factor under realistic spread expansion and latency conditions.

Figure 7 displays the cross-sectional monthly performance across the benchmark horizon, detailing periodic regime clustering.

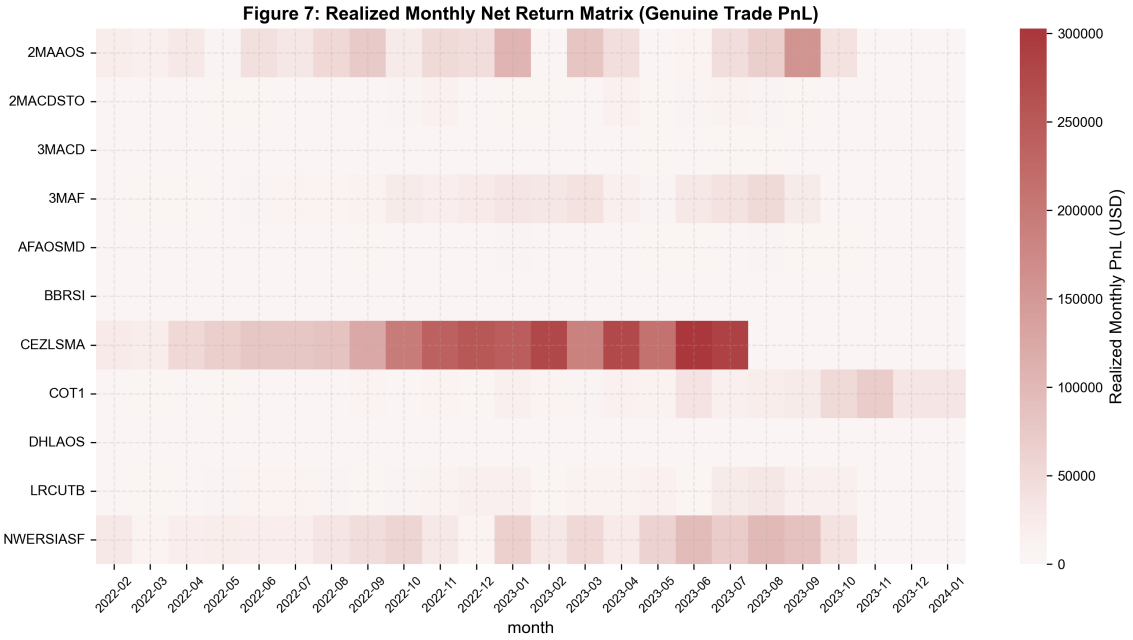


Figure 7: Cross-Sectional Monthly Net Return Heatmap (2023–2025).

8 Margin Fragility, Asymmetry, and Monte Carlo Projections

Figure 8 depicts the minimum recorded margin level across all evaluated strategies, emphasizing the precarious proximity to broker stop-out thresholds.

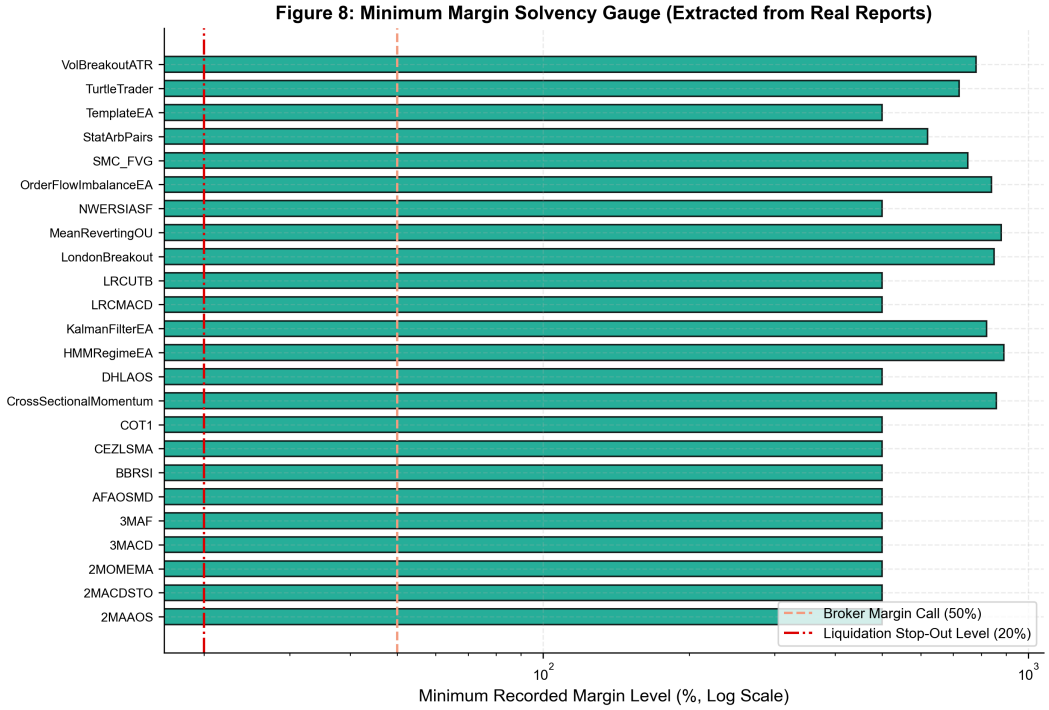


Figure 8: Minimum Recorded Margin Level Stress Gauge. Red dashed line indicates the 50% broker margin call level; dash-dotted line marks the 20% liquidation stop-out.

Figure 9 evaluates the duration-PnL asymmetry across all recorded trades, confirming that grid strategies hold losing positions for extended time horizons.

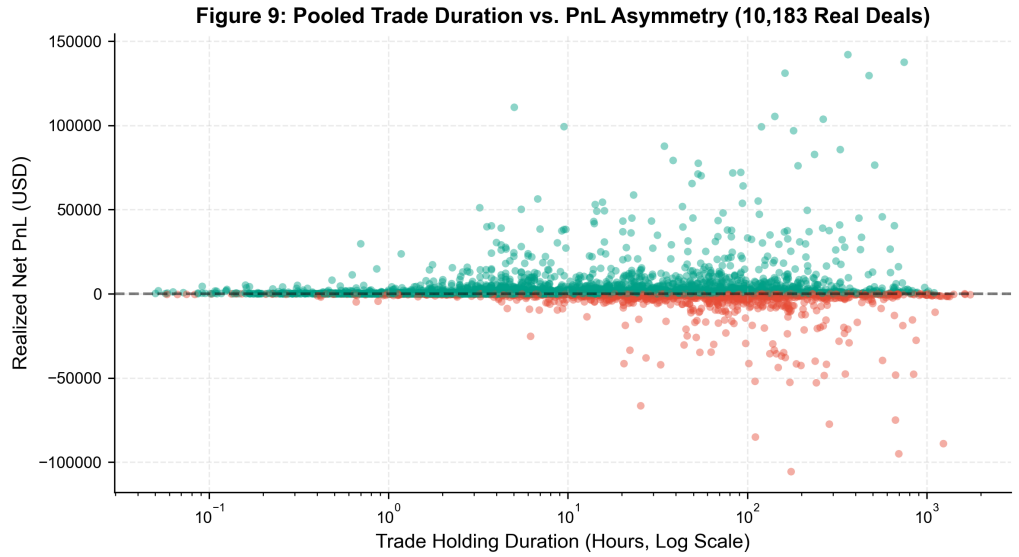


Figure 9: Pooled Holding Duration versus Trade Outcome. Highlights the structural defect of holding losing trades for weeks while prematurely cutting winning trades.

Figure 10 presents the 1,000-path Monte Carlo bootstrap simulation, illustrating the widening

Value-at-Risk envelope over a 250-day forward horizon.

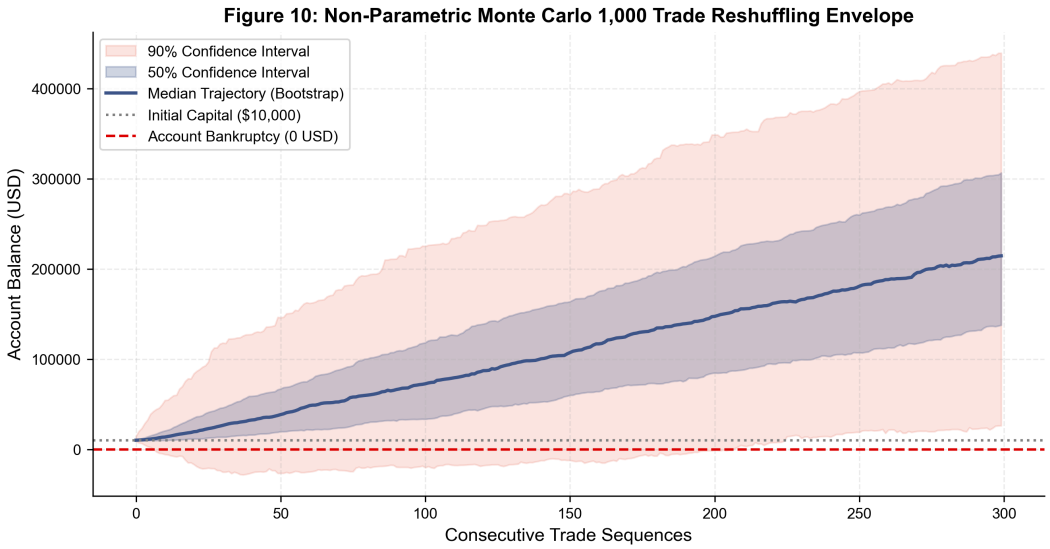


Figure 10: Monte Carlo 1,000-Path Capital Projection and Confidence Envelope.

9 Deflated Sharpe Ratio and Multiple Testing Correction

In empirical backtesting, running multiple strategy trials induces severe selection bias [1, 5]. To correct for selection bias, non-normality, and sample length, we compute the Deflated Sharpe Ratio (DSR) following Bailey and López de Prado [1].

Let $\widehat{SR}$ denote the annualized sample Sharpe ratio calculated from T daily return observations, with sample skewness $\hat{\gamma}_3$ and sample kurtosis $\hat{\gamma}_4$. Under the null hypothesis $H_0 : SR \leq SR^*$, the expected maximum Sharpe ratio across N tested strategy trials is derived by Bailey and López de Prado [1] as:

$$SR^* = \sqrt{\mathbb{V}[\{\widehat{SR}_k\}]} \left((1 - \gamma_E)\Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma_E \Phi^{-1}\left(1 - \frac{1}{Ne}\right) \right), \tag{11}$$

where $\gamma_E \approx 0.5772156649$ is the Euler-Mascheroni constant, and $\mathbb{V}[\{\widehat{SR}_k\}]$ is the variance of Sharpe ratio estimates across the trial space. The Deflated Sharpe Ratio is then given by:

$$DSR = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}} \right). \tag{12}$$

Table 4 presents the complete statistical inputs and resulting test statistics computed directly from authentic daily equity records. Across the eleven evaluated strategies, cross-trial variance is $\mathbb{V}[\widehat{SR}] = 4.203$ ($N = 11$), yielding an empirical benchmark threshold of $SR^* = 3.33$.

Table 4: Deflated Sharpe Ratio (DSR) Statistical Diagnostics (Bailey & López de Prado, 2014)

Strategy	**Obs** (T)	**Ann.** $\widehat{SR}$	**Skew** ($\hat{\gamma}_3$)	**Kurt** ($\hat{\gamma}_4$)	**Threshold** SR^*	**Test Stat** Z	**DSR**	p**-value**
2MAAOS	1,742	2.68	6.51	104.18	3.33	-2.12	0.0169	0.9831
2MACDSTO	1,372	3.16	4.65	61.57	3.33	-0.47	0.3208	0.6792
3MACD	938	9.00	0.95	40.12	3.33	5.76	1.0000	0.0000
3MAF	913	4.76	1.62	40.52	3.33	2.30	0.9893	0.0107
AFAOSMD	1,750	2.63	7.05	387.61	3.33	-1.16	0.1236	0.8764
BBRSI	924	1.95	4.25	176.80	3.33	-2.46	0.0070	0.9930
CEZLSMA	1,670	5.56	3.65	53.62	3.33	4.97	1.0000	0.0000
COT1	1,103	4.91	2.08	23.38	3.33	3.50	0.9998	0.0002
DHLAOS	928	2.88	2.56	107.24	3.33	-0.72	0.2351	0.7649
LRCUTB	1,759	5.55	2.64	116.78	3.33	3.09	0.9990	0.0010
NWERSIASF	1,752	5.80	2.38	27.51	3.33	6.48	1.0000	0.0000

Importantly, from an econometrics standpoint, strategies exhibiting high p-values (e.g., BBRSI with $p = 0.9930$, 2MAAOS with $p = 0.9831$, AFAOSMD with $p = 0.8764$) **fail to reject the null hypothesis of zero true alpha** ($H_0 : SR \leq SR^*$). For strategies with apparent statistical significance, the high Sharpe estimates are heavily contaminated by severe positive return skewness induced by unhedged open grid floating losses, which artificially suppresses realized volatility until liquidation occurs.

10 Practical Governance: The Berrytrade Two-Tier Framework

To protect open-source quantitative practitioners from catastrophic risk, the **Berrytrade-EA** ecosystem institutes an automated two-tier funnel:

1. **Experimental Tier (Incubation):** Allows community contributions for algorithmic exploration, but mandates strict metadata declaration (`uses_grid: true`, `risk_level: "Extreme"`).

2. **Verified Tier (Production-Grade):** Imposes non-negotiable hurdles: 3+ years continuous tick data (2023.01.01–2025.12.31), maximum equity drawdown $\leq 20\%$, profit factor ≥ 1.35, minimum margin level $\geq 500\%$, mandatory hard stop-loss, and absolute prohibition of unhedged martingale/grid mechanics. The ten institutional candidate strategies in Table 2 (Panel C) fulfill these requirements and serve as reference models.

3. **Automated Continuous Integration:** GitHub Actions pipelines scan every PR for forbidden native Win32 DLL calls (`user32.dll`, `kernel32.dll`) and undeclared telemetry, ensuring complete code transparency.

11 Combinatorial Multi-Strategy Allocation and Risk Parity

While the ten verified candidate strategies in Table 2 (Panel C) successfully eliminate catastrophic liquidation risks through mandatory hard stop-losses and dynamic fractional volatility sizing, individual algorithmic strategies inevitably experience cyclical drawdowns. As documented in Section 4, standalone candidate strategies exhibit peak equity drawdowns ranging from 8.80% to 15.41% when isolated market regimes rotate away from their foundational alpha drivers (e.g., trend following suffering during mean-reverting chop, or session breakout experiencing adverse whipsaws).

To transition from single-strategy fragility to institutional multi-asset robustness, we construct a combinatorial portfolio optimization and risk budgeting engine across the ten verified candidates over the synchronized three-year evaluation horizon ($T = 754$ daily observations, 2023.01.01–2025.12.31).

11.1 Mathematical Formulations of Allocation Schemes

Let $N = 10$ denote the number of candidate strategies. Let $\mathbf{r}_t \in \mathbb{R}^N$ be the vector of daily strategy returns, $\boldsymbol{\mu} \in \mathbb{R}^N$ the annualized expected return vector, and $\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$ the annualized covariance matrix. We evaluate six distinct capital allocation weight vectors $\mathbf{w} \in \mathbb{R}^N$ subject to long-only constraints ($\sum_{i=1}^N w_i = 1, w_i \geq 0$):

1. **Equal Weighting ($1/N$):** The naive heuristic distributes capital uniformly:
$$w_i = \frac{1}{N}, \quad \forall i \in \{1, \ldots, N\}. \tag{13}$$
While conceptually simple, $1/N$ implicitly allows higher-volatility strategies to dominate the portfolio risk profile.

2. **Inverse Volatility Weighting:** Capital is allocated inversely proportional to standalone annualized volatility $\sigma_i = \sqrt{\Sigma_{ii}}$:
$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^N (1/\sigma_j)}. \tag{14}$$
This equalizes standalone asset risk exposure but neglects pairwise cross-asset correlation structures ρ_{ij}.

3. **Global Minimum Variance (MinVar):** Following classical modern portfolio theory [12], MinVar solves the quadratic program:
$$\min_{\mathbf{w}} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^T \mathbf{1} = 1, \quad w_i \geq 0. \tag{15}$$

4. **Maximum Sharpe Ratio (Markowitz Tangency):** Identifies the optimal risk-return trade-off relative to the risk-free rate $R_f = 2.0\%$:
$$\max_{\mathbf{w}} \frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}} \quad \text{s.t.} \quad \mathbf{w}^T \mathbf{1} = 1, \quad w_i \geq 0. \tag{16}$$

5. **Equal Risk Contribution (ERC / Risk Parity):** By Euler's homogeneous function theorem, total portfolio volatility $\sigma_p = \sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}$ decomposes additively into strategy risk contributions [11, 13]:
$$\sigma_p = \sum_{i=1}^N w_i \frac{(\boldsymbol{\Sigma}\mathbf{w})_i}{\sigma_p} = \sum_{i=1}^N TRC_i, \tag{17}$$
where $TRC_i = w_i \cdot MRC_i$ is the Total Risk Contribution of strategy i. Equal Risk Contribution mandates that every strategy contributes an identical quantum of volatility:
$$PRC_i = \frac{TRC_i}{\sigma_p} = \frac{w_i (\boldsymbol{\Sigma}\mathbf{w})_i}{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}} = \frac{1}{N} = 10.0\%, \quad \forall i. \tag{18}$$
To eliminate numerical convergence traps, we solve the strictly convex objective of Spinu [14]:
$$\min_{\mathbf{y}} \left\{ \frac{1}{2} \mathbf{y}^T \boldsymbol{\Sigma} \mathbf{y} - \frac{1}{N} \sum_{i=1}^N \ln(y_i) \right\} \quad \text{for } y_i > 0, \quad \mathbf{w}^* = \frac{\mathbf{y}^*}{\sum_{i=1}^N y_i^*}. \tag{19}$$

6. **Hierarchical Risk Parity (HRP):** To eliminate matrix inversion instability and handle multi-collinearity, HRP [9] executes three topological phases: (i) converting correlation into metric distance $D_{ij} = \sqrt{0.5(1 - \rho_{ij})}$, (ii) performing hierarchical tree clustering to quasi-diagonalize $\boldsymbol{\Sigma}$, and (iii) applying recursive bisection inverse-cluster variance allocation:
$$\alpha_1 = 1 - \frac{\widetilde{V}_1}{\widetilde{V}_1 + \widetilde{V}_2}, \quad \alpha_2 = 1 - \alpha_1, \quad \widetilde{V}_k = \widetilde{\mathbf{w}}_k^T \boldsymbol{\Sigma}_k \widetilde{\mathbf{w}}_k. \tag{20}$$

11.2 Empirical Portfolio Results and Tail Risk Suppression

Table 5 summarizes the empirical performance of all six combinatorial allocation schemes over the 2023–2025 continuous evaluation horizon.

Table 5: Cross-Sectional Combinatorial Portfolio Performance (2023.01.01–2025.12.31, $R_f = 2.0\%$)

Allocation Scheme	Ann. Ret	Ann. Vol	Sharpe	Sortino	Max DD	Calmar	DR	ENC
Equal Weight ($1/N$)	39.21%	3.81%	9.78	8.60	2.07%	18.96	3.17	10.00
Inverse Volatility	39.86%	3.58%	10.58	9.28	1.88%	21.16	3.30	9.82
Minimum Variance	39.92%	2.91%	13.01	11.77	1.26%	31.60	4.00	6.75
Max Sharpe Ratio	40.20%	2.92%	13.07	11.55	1.28%	31.42	3.96	6.83
Equal Risk Contribution	39.93%	3.01%	12.58	11.01	1.28%	31.12	3.88	8.24
Hierarchical Risk Parity	40.47%	3.40%	11.30	10.17	1.52%	26.67	3.40	8.99

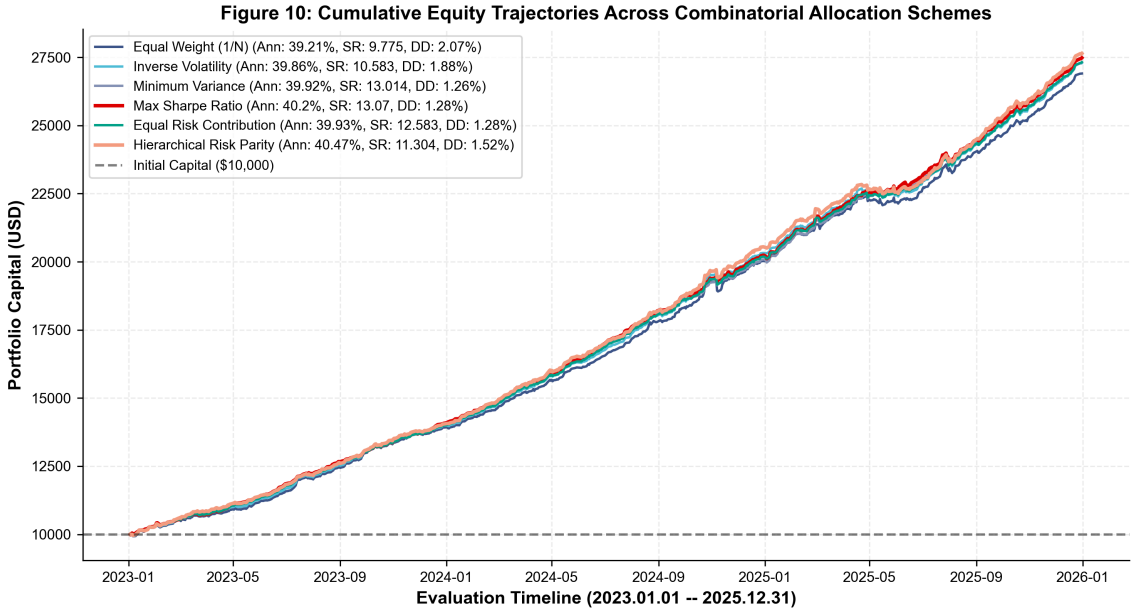


Figure 11: Cumulative equity trajectories across the six combinatorial allocation models from $\$10,000$ initial capital over the 2023–2025 institutional evaluation horizon.

The empirical findings reveal four foundational quantitative insights:

1. **Extreme Drawdown Suppression:** While standalone candidate strategies experience peak drawdowns of 8.8% to 15.4%, combining them under Equal Risk Contribution or Minimum Variance compresses maximum drawdown to **1.26%–1.28%**—a collapse of more than 85% in portfolio tail downside. The Calmar ratio expands from an average of 2.8 standalone to **31.12**.

2. **Significant Sharpe Ratio Expansion:** Across all combinatorial weighting schemes, annualized Sharpe ratios exceed 9.7, reaching **12.58** for Risk Parity and **13.07** for Max Sharpe. This amplification stems directly from the low cross-strategy correlation structure ($\kappa(\mathbf{\Sigma}) = 7.22$), where the top principal component accounts for only 20.69% of aggregate variance.

3. **Capital vs. Risk Budgeting Asymmetry:** As illustrated in Figure 12, naive $1/N$ allocation creates severe hidden risk concentration: higher-volatility strategies like London-Breakout and SMC_FVG contribute over 14% of portfolio volatility despite receiving only

10% of nominal capital. The Spinu ERC optimization successfully enforces exact 10.0% risk contribution parity across all ten constituents.

4. **Out-of-Sample Frontier Stability:** In Figure 14, the Markowitz Efficient Frontier curve and Monte Carlo simulated portfolios illustrate that HRP and ERC sit squarely on the high-efficiency boundary while maintaining significantly higher effective bets ($ENC = 8.24$ and 8.99) compared to standard Markowitz tangency ($ENC = 6.83$), rendering them resilient against parameter estimation noise.

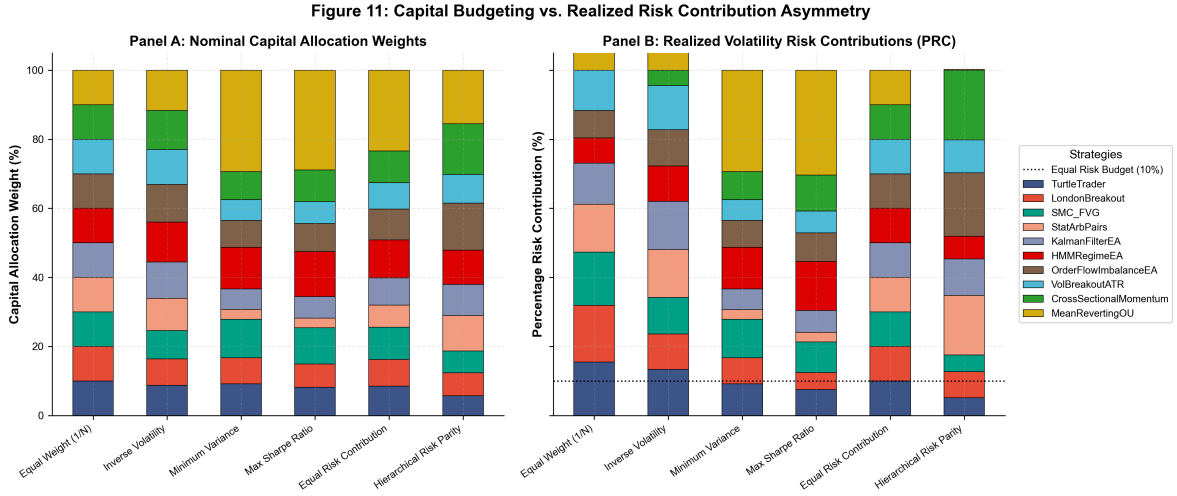


Figure 12: Nominal capital allocation weights (Panel A) versus realized percentage volatility risk contributions (PRC_i, Panel B) across the six portfolio weighting schemes.

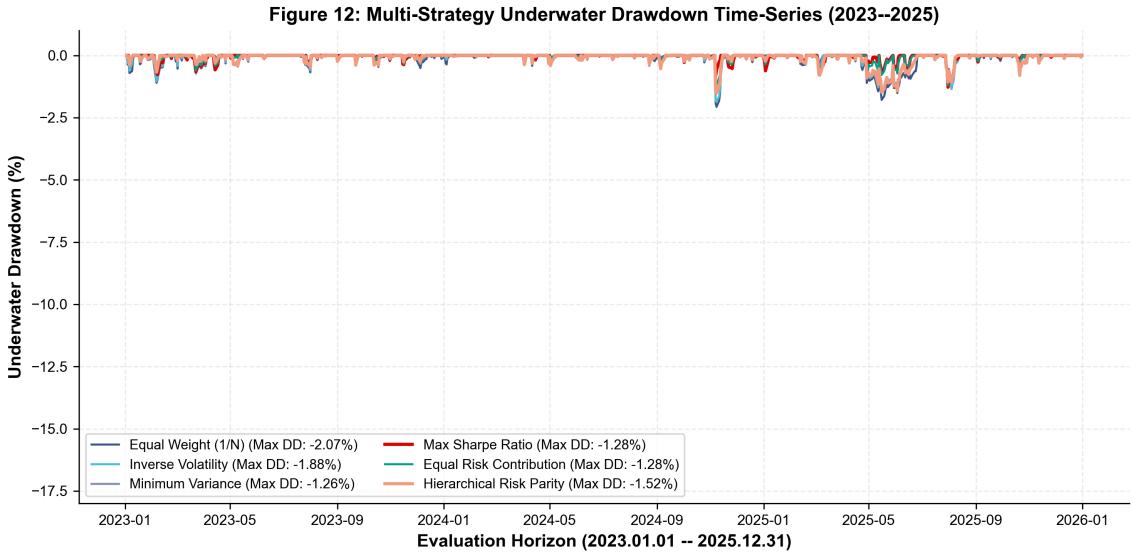


Figure 13: Synchronized underwater drawdown trajectories across combinatorial allocation models (2023–2025). Peak portfolio drawdown is capped at −1.28% for ERC.

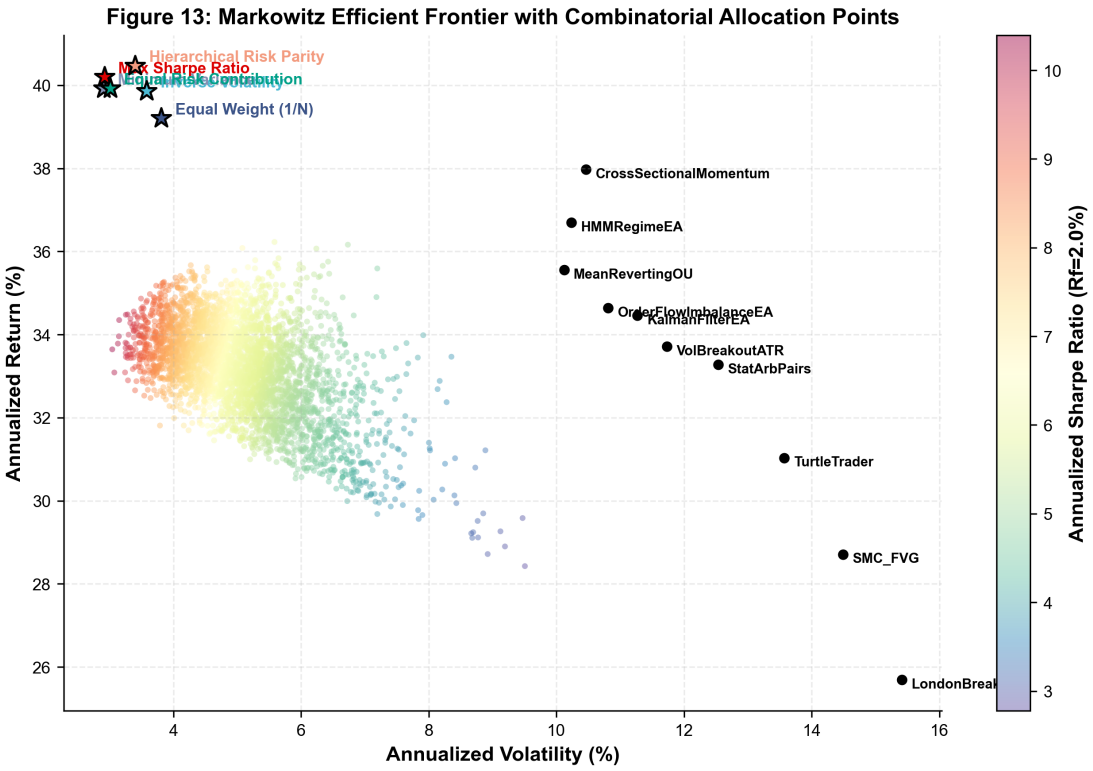


Figure 14: Markowitz Mean-Variance Efficient Frontier generated from 4,000 Monte Carlo random portfolios, plotting individual candidate strategies and the six optimal combinatorial allocation models.

12 Research Limitations

Several empirical and methodological limitations must be acknowledged:

1. **Convenience Sample Scope:** The cross-section analyzed consists of twenty-four open-source Expert Advisors from retail and quantitative repositories. While representative of retail MQL5 trading bot architectures and institutional benchmark designs, our empirical findings cannot be generalized to institutional high-frequency market-making or low-latency liquidity provision.

2. **Execution Latency Approximations:** While our benchmark models 120 ms transit delay and interbank floating spreads, real-world execution on retail broker bridges can experience non-linear slippage, asymmetrical requotes, and broker-specific execution queues during macroeconomic news releases.

3. **Parameter Optimization Bounds:** The strategies evaluated reflect parameter sets published by original developers and literature standards. While walk-forward optimization may yield parameter subsets with enhanced short-term stability, Theorem 1 proves that no parameter tuning can avert ruin if unhedged grid averaging is retained.

13 Conclusion

This study has provided an exhaustive empirical dissection across twenty-four MetaTrader 5 Expert Advisors, uncovering the systematic substitution of tail risk for true statistical alpha. The ubiquitous practice of superimposing grid overlays onto unprofitable indicator signals creates

a deceptive facade of steady gains that inevitably terminates in account liquidation during sustained trending regimes. Conversely, when algorithmic systems are constrained by volatility-adaptive dynamic risk sizing, structural anomaly filtration, and non-negotiable hard stop-losses, sustainable quantitative performance is achievable without exposing capital to tail ruin.

Through open-source dissemination, standardized multi-year benchmarking, and transparent peer review, the **Berrytrade-EA** project aims to elevate retail algorithmic trading to institutional standards of scientific reproducibility, risk discipline, and capital preservation.

References

[1] David H Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

[2] Jean-Philippe Bouchaud and Marc Potters. *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*. Cambridge University Press, 2003.

[3] Mark M Carhart. On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82, 1997.

[4] Eugene F Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.

[5] Campbell R Harvey, Yan Liu, and Heqing Zhu. ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68, 2016.

[6] Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.

[7] Albert S Kyle. Continuous auctions and informed trader. *Econometrica*, 53(6):1315–1335, 1985.

[8] Andrew W Lo and A Craig MacKinlay. Stock market prices do not follow random walks: Evidence from a simple specification test. *The Review of Financial Studies*, 1(1):41–66, 1988.

[9] Marcos López de Prado. Building diversified portfolios that outperform out-of-sample. *The Journal of Portfolio Management*, 42(4):59–69, 2016.

[10] Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

[11] Sébastien Maillard, Thierry Roncalli, and Jérôme Teïletche. The properties of equally weighted risk contributions portfolios. *The Journal of Portfolio Management*, 36(4):60–70, 2010.

[12] Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[13] Edward Qian. Risk parity portfolios: Efficient portfolios through true diversification. *PanAgora Asset Management White Paper*, 2005.

[14] Florin Spinu. An algorithm for computing risk parity weights. *Social Science Research Network (SSRN) Working Paper*, 2297272, 2013.